

Work and Practical Session: Machine Learning

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Session 02: Supervised Learning – Artificial Neural Networks

- Work session

Exercise #1: Binary classification

We assume a binary classification problem with **one** feature x in real space.

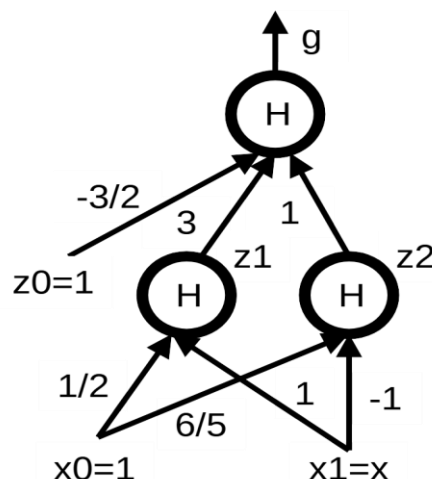
We have eight 1-dimensional inputs and eight corresponding targets given by:

$X =$	$T =$
-2	0
-1.5	0
-1	0
-0.5	1
0	1
0.5	1
1	0
1.5	0

The purpose here is to find a neural network that will classify correctly this dataset.

1. Draw this datapoints in 2D using the black color for class 0 and the red color for class 1.
2. Are these 1-dimensional data linearly separable?

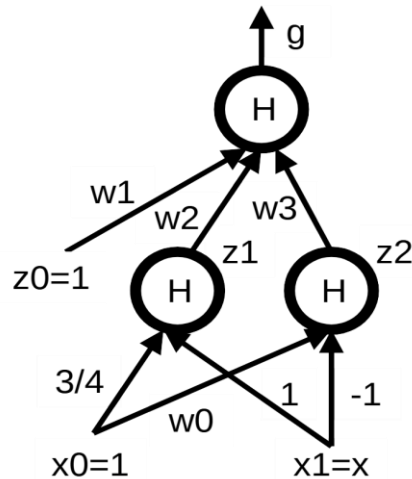
We wish to use an artificial neural network to classify our inputs. In this artificial neural network, all the activation functions are the Heaviside functions H , where $H(x)=1$ when $x \geq 0$ and $H(x)=0$ when $x < 0$. Consider the following multi-layer perceptron:



3. z_1 and z_2 are the outputs of the two neurons in the first hidden layer. Represent the data in the plane (z_1, z_2) once it has passed through the first layer.
4. Describe the function implemented by this neural network.

5. Give the confusion matrix of this model and compute the error rate.

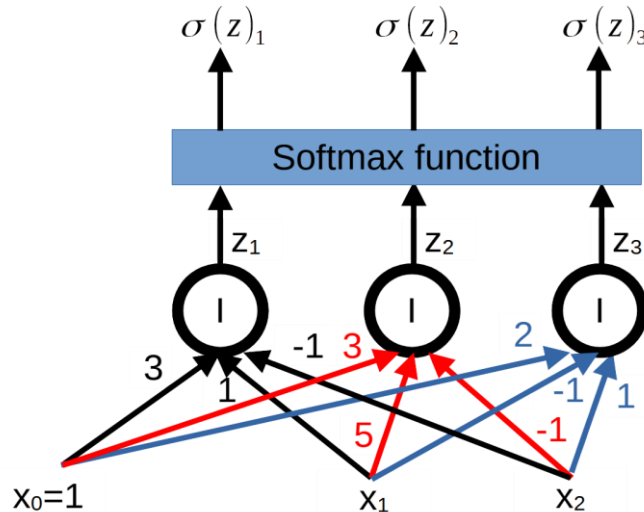
The previous model makes some errors and we would like to reduce them by proposing another one.



6. From this figure, determine the synaptic value w_0 such that the datapoints (X, T) are linearly separable in the plane (z_1, z_2) .
7. In the second layer, determine the synaptic coefficients (several values are also possible) w_1 , w_2 and w_3 , in order to reach a zero error rate.

Exercise #2 : 3-class Classification

Consider a classification problem where the outputs $z = (z_1, z_2, z_3)$ are linear combinations of the inputs and weights, where x_1 and x_2 are the real input variables and $x_0=1$. We have obtained the following weights of an artificial neural network from a dataset :



The activation function of the output layer is *softmax*. The softmax function can be used for a multi-class classification problem. It returns an output vector of K real positive values whose sum equals 1, from a given vector $z=(z_1, z_2, \dots, z_K)$. The j^{th} component of this output vector is given by:

$$\sigma(\mathbf{z})_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}}$$

This gives the probability that the input belongs to class j .

Assuming the input vector $(x_0, x_1, x_2)=(1, -1, -2)$, determine the values z_1 , z_2 and z_3 obtained by the

neurons, as well as the prediction results $\sigma(z)$ given by the softmax function.

- Practical session

You can load your CSV data using NumPy and the `numpy.loadtxt()`.

This function assumes there is no header in the first row of the file and all data has the same format. Here too, data must be numeric. The example below assumes that the file 'pima-indians-diabetes.data.csv' is in your current working directory.

```
import numpy  
filename='pima indians diabetes.data.csv'  
raw_data=open(filename, 'rt')  
data =numpy.loadtxt(raw_data, delimiter=',')  
print(data,data.shape)
```

Exercise #1 :

Study the python code template `_ANN.py` where an ANN is applied on the Churn Modelling dataset. The Churn Modelling dataset refers to customers who has exited a bank or not within the next 6 months from where the data was collected.

It contains 10000 instances and 12 relevant features (CreditScore, Geography, Age, Tenure, EstimatedSalary...).

The architecture that is used for the neural network is: ((6,'relu'),(6,'relu'),(1,'sigmoid'))

With the trained ANN, predict the result for this person:

Geography: France

Credit Score: 600

Gender: Male

Age: 40

Tenure: 3

Balance: 60000

Number of Products: 2

Has Credit Card: Yes

Is Active Member: Yes

Estimated Salary: 50000

Exercise #2 :

Implement an ANN and train it on the Iris Plant dataset using the first 100 examples (corresponding to classes Iris-setosa and Iris-versicolor) and using as input features the sepal length (first column) and the petal length (third column).

The Iris Plant dataset is perhaps the best known database to be found in the pattern recognition literature. The data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are NOT linearly separable from each other. The attribute to predict is the class of iris plant.

Attribute Information:

1. sepal length in cm

2. sepal width in cm

3. petal length in cm

4. petal width in cm

5. class: Iris Setosa, Iris Versicolour or Iris Virginica